Patterns vs Rules vs Analogies

Patterns; recurring structure or arrangements of linguistic entity in language

Structural Identity,

Arrangement of sound units

Arrangement of morphemes in words

Arrangement of words in sentences

Phonological Pattern

- How speech sounds are organized, combined, and used to distinguish meaning.
  - /θ/ as in *think*
  - /ð/ as in *this*
  - /v/, /z/, /ʒ/ (*measure*)
- Does **not** contrast aspirated and unaspirated stops as separate phonemes.
- **Example**
- *pin* /pʰɪn/ and *spin* /spɪn/
  - Aspiration changes according to context but does **not** change meaning.
  - Bus-buses, cat-cats, bench, benches, Dog-dogs

Stress-proDuct, Production, reCord, Record

Type	Example
voiceless unaspirated	/k/ कल *kal*
voiceless aspirated	/k^{h}/ खल *khal*

Syntactic Pattern

- English SVO John ate the apple.
- Hindi SOV जॉन ने सेब खाया। (Jān ne seb khāyā.)
- Verb –complement
- Specifier-verb-complement

Morphology-Light Verb construction

- V1 , V2 construction
- राम ने खाना खा लिया। Ram ERG food eat take.PSTRam finished eating the food. Completion
- उसने किताब पढ़ डाली। he book read put.PST He read the book completely. Exhaustive completion
- बच्चा रो पड़ा। child cry fall.PST The child burst into tears. Sudden inception
- वह हँस पड़ा। he laugh fall.PST He suddenly started laughing. Sudden beginning

Rules

- the systematic principles that govern how sounds, words, phrases, and sentences are formed and interpreted. These rules are not arbitrary prescriptions ("don't split infinitives"), but rather the **implicit patterns that native speakers unconsciously know and use** to produce and understand language.

Rules

- A **rule** is a more explicit formulation of a linguistic regularity. It specifies **what operation should apply under particular conditions**.

Analogy; it creates abstract meaning

- **Analogy** is the process by which a speaker extends a familiar pattern from one linguistic item to another **because the two are structurally or functionally similar**.
- For example, suppose a child knows:
- sing → sang
- Walk-walked
- Dog—dogs. Child-childs
- and encounters a new or less familiar verb. The child may use an existing morphological pattern to construct its form.

- Thus analogy is fundamentally a **mapping operation**.
- A:B::C:X
- which means:
- **A is to B as C is to X.**
- For example:
- walk : walked :: jump : jumped
- The speaker maps the relationship:
- VERB→VERB + ed
- onto another lexical item.

- We can conceptualize them as:
- Pattern→Rule→Analogy
- But this is not necessarily a one-way historical sequence. In actual language use, the relationship is **dynamic**:
- Experience→Pattern recognition→Rule abstraction→Analogical extension→New experience→Pattern

- For example:
- **Input**
- walk → walked
  talk → talked
  play → played
- ↓
- **Pattern recognition**
- verbs frequently form past tense with **-ed**
- ↓
- **Rule abstraction**
- V→V+ed
- ↓
- **Analogy**
- jump → jumped
- ↓
- **New pattern**
- The new example reinforces the original generalization.

Key Features of Analogy:

- **Paradigm Regularization**:
  - Irregular forms are changed to align with a more common pattern.
  - Example: In English, the irregular past tense of *help* was once *holp*, but analogy with regular verbs led to the now-standard *helped*.
- **Proportional Analogy**:
  - This involves creating a new form based on an existing pattern. It can be represented as:
    - A : B = C : X
    - For instance, *sing* : *sang* = *bring* : *brang* (although *brang* is non-standard, it illustrates how analogy can operate).

- **Leveling**:
  - Differences within a morphological paradigm are reduced or eliminated.
  - Example: In Latin, the word for "man" had various forms (*homo, hominis*), but in some Romance languages (e.g., Italian *uomo*), analogy reduced the paradigm's irregularities.
- **Extension**:
  - A pattern or rule is extended to new contexts.
  - Example: In English, the spread of the -s plural marker to words that historically used other plural forms (e.g., *children* instead of older *childer*).

- **Types of Analogy:**
- **Analogical Change**: The introduction of new linguistic forms based on patterns of existing ones.
- **Reanalysis**: Misinterpreting a structure or pattern, leading to a new linguistic form.
- **Analogy and Borrowing**: Loanwords often undergo analogical adjustments to fit the phonological or grammatical system of the borrowing language.